

RetinoPATH

Automated Diabetic Retinopathy Grading Using RETFound-DINOv2

Project Progress Report | 19 June 2026

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Institution	Research Project — UK NHS Fundus Cohort
Model	RETFound-DINOv2-MEH (ViT-Large, 307M parameters)
Task	4-Class Diabetic Retinopathy Grading (R0 / R1 / R2 / R3A)
Report Date	19 June 2026
Status	Phase 2B Complete — Patient Aggregation Complete — R3A Analysis Complete

Key Results at a Glance

Metric	Baseline (P1)	Best Achieved (P2B + PtMean)	Improvement
AUROC	0.826	0.946	+12 pp
Cohen's Kappa	0.667	0.821	+15 pp
R0 Sensitivity	0.905	0.989	+8 pp
R1 Sensitivity	0.390	0.810	+42 pp
R2 Sensitivity	0.510	0.583	+7 pp
R3A Sensitivity	0.417	0.458	+4 pp *

* R3A (PDR) improvement is modest due to severe class sparsity (9 test patients, 43 training patients). Best image-level R3A sensitivity = 0.458 (P2B + Youden). See Section 6 for full analysis.

1. Executive Summary

This report documents the end-to-end development of an automated diabetic retinopathy (DR) grading system built on the RETFound-DINOv2-MEH foundation model, fine-tuned on a UK NHS fundus imaging cohort. The project progressed through seven distinct stages, from raw data pipeline construction to a fully fine-tuned model with patient-level aggregated predictions.

The primary objective was to maximise per-class sensitivity for four DR grades (R0: normal, R1: mild/moderate NPDR, R2: moderate/severe NPDR, R3A: PDR), with particular clinical emphasis on R1 detection — the earliest actionable stage for treatment referral. The final best configuration (Phase 2B full fine-tune + patient-level mean aggregation + argmax decision) achieved:

- **AUROC 0.9456** — up from 0.826 at baseline (+12 percentage points)
- **Cohen's Kappa 0.8212** — first configuration to cross the 0.80 clinical agreement threshold
- **R1 sensitivity 0.8095** — up from 0.390 at baseline (+42 pp)
- **R0 sensitivity 0.9890** — near-perfect normal detection

The remaining open challenge is R3A (PDR) sensitivity, which is constrained by the small number of PDR cases in the dataset (43 training patients, 9 test patients). A dedicated threshold optimisation sweep confirmed this is a data limitation rather than a modelling limitation.

2. Project Overview

2.1 Clinical Context

Diabetic retinopathy is the leading cause of preventable blindness in working-age adults. Early detection (R1 stage) enables timely treatment before irreversible vision loss occurs. Automated grading can dramatically expand the reach of screening programmes by reducing the burden on specialist reading centres.

2.2 Backbone Model

RETFound-DINOv2-MEH is a Vision Transformer (ViT-Large) pretrained specifically on fundus photographs from Moorfields Eye Hospital (MEH), London, using the DINOv2 self-supervised learning framework. With 307 million parameters across 24 transformer blocks, it provides rich domain-specific representations of retinal structure — vessel morphology, optic disc features, and macular patterns — without requiring ImageNet-pretrained weights that would be suboptimal for fundus images.

2.3 Task Definition

Grade	Description	Clinical Action
R0	No DR / Normal	Reference class — no referral required
R1	Mild / Moderate NPDR	Earliest actionable stage — annual surveillance or referral
R2	Moderate / Severe NPDR	Urgent ophthalmology referral
R3A	Proliferative DR (PDR)	Emergency referral — high risk of vision loss

3. Dataset

3.1 Source and Structure

The dataset comprises fundus photographs from a UK NHS diabetic retinopathy screening cohort. Images were graded by a reading centre using a structured hierarchy: Arbitration grade (when present) supersedes Secondary grade, which supersedes Primary grade. Only ranking=1 rows (the definitive final grade per patient) were used.

3.2 Dataset Statistics

Statistic	Value
Total unique patients	2,401
Empty eye-folders (excluded)	432
Images used in CV + test	4,777 (4,075 train/val + 702 test)
Patients in CV set	990
Patients in test set	175
Typical images per patient	2 (some 3–8)

3.3 Class Distribution

Grade	CV Patients	Test Patients	CV %	Test %
R0 (Normal)	518 / 990	91 / 175	52.3%	52.0%
R1 (Mild/Mod NPDR)	362 / 990	63 / 175	36.6%	36.0%
R2 (Mod/Severe NPDR)	67 / 990	12 / 175	6.8%	6.9%
R3A (PDR)	43 / 990	9 / 175	4.3%	5.1%

The dataset is moderately imbalanced. R3A is severely under-represented (4.3% of training patients), which is the root cause of the R3A sensitivity ceiling observed throughout all training phases.

3.4 Data Split Strategy

A stratified patient-level split was used: all images from a given patient appear in *exactly one* of train, validation, or test. This prevents data leakage that would occur if the same patient appeared in both training and evaluation sets. The 5-fold cross-validation used the same patient-level stratification, ensuring that out-of-fold predictions are genuinely out-of-distribution.

4. Training Phases

4.1 Phase 1 — 5-Fold Cross-Validation, Linear Probe (CE Loss)

The first training phase established the baseline by fine-tuning only the **classification head** (a linear layer mapping the 1024-dim ViT-Large CLS token to 4 DR grades), while keeping all 307M backbone parameters frozen. This approach, called a "linear probe," tests how well the pretrained representations already encode DR-relevant features without any backbone adaptation.

Configuration

Setting	Value
Backbone parameters	Frozen (not trained)
Trained parameters	Classification head only (~4K params)
Loss function	Cross-Entropy with class weights (inverse frequency)
Optimiser	AdamW
Learning rate	1e-3 (head only)
Batch size	32
Cross-validation	5-fold stratified, patient-level
Input size	224 × 224

Phase 1 Test Set Results (Image Level)

Metric	Value
AUROC	0.8259
Cohen's Kappa	0.6665
Accuracy	0.6866
Macro Sensitivity	0.5554
R0 Sensitivity	0.9054
R1 Sensitivity	0.3898 ← primary weakness
R2 Sensitivity	0.5098
R3A Sensitivity	0.4167

Key finding: The frozen backbone already achieves reasonable overall performance (AUROC 0.826), confirming that the RETFound-DINOv2 representations transfer well to this UK NHS dataset. However, R1 sensitivity of 0.39 is clinically insufficient — the model missed more than 60% of early-stage DR cases.

4.2 Phase 2C — Per-Class Youden Threshold Tuning

After Phase 1 cross-validation, out-of-fold (OOF) probability predictions were used to calibrate **per-class decision thresholds** using the Youden J statistic ($J = \text{sensitivity} + \text{specificity} - 1$). The default argmax decision (pick the highest probability class) does not account for class imbalance; Youden thresholds correct for this by assigning a class-specific probability threshold above which a class is considered "detected".

Decision Rule

For a prediction vector $p = [p_{R0}, p_{R1}, p_{R2}, p_{R3A}]$ and threshold vector $t = [t_{R0}, t_{R1}, t_{R2}, t_{R3A}]$, the OvR (one-vs-rest) ratio rule is:

```
predicted class = argmax(p / t) if any p_i > t_i, else argmax(p)
```

Tuned Thresholds (Phase 1 OOF)

Class	Youden Threshold
R0	0.4907
R1	0.3160
R2	0.1252
R3A	0.1463

Finding: Youden thresholds improved R2 sensitivity (by lowering its decision bar) but slightly reduced R1 sensitivity and overall kappa at image level. The thresholds optimised at image level do not transfer reliably to patient-level aggregated probabilities, a pattern that persisted across all subsequent phases.

4.3 Phase 2A — Focal Loss (Linear Probe)

Phase 2A repeated the linear probe cross-validation but replaced cross-entropy loss with **focal loss** ($\gamma = 2$). Focal loss down-weights the contribution of easy-to-classify examples (high confidence, correct predictions) and concentrates training signal on hard examples (low confidence or misclassified). The formula is:

$$FL = -\alpha \cdot (1 - p_{\text{true}})^{\gamma} \cdot \log(p_{\text{true}})$$

where p_{true} is the predicted probability for the correct class and $\gamma = 2$ means a correctly predicted probability of 0.9 receives only 1% of the loss weight it would under standard cross-entropy. The intent was to push the model harder on ambiguous R1/R2 cases.

Phase 2A Test Set Results vs Phase 1

Metric	Phase 1	Phase 2A	Δ
AUROC	0.8259	0.8180	−0.008
Cohen's Kappa	0.6665	0.6491	−0.017
R1 Sensitivity	0.3898	0.5042	+0.114
R3A Sensitivity	0.4167	0.2917	−0.125

Finding: Focal loss improved R1 sensitivity (+11 pp) but at cost to AUROC and R3A. The net effect at image level was slightly negative for overall agreement. However, patient-level aggregation recovered kappa to 0.770, matching Phase 1. Focal loss alone (with frozen backbone) is insufficient to close the R1 gap.

4.4 Phase 2B — Full Fine-Tuning (All 307M Parameters)

Phase 2B unfroze the entire ViT-Large backbone, training all 307 million parameters jointly with the classification head. This required three memory management techniques to fit within the 12 GB GPU (RTX 3060):

- **Layer-Wise Learning Rate Decay (LLRD, decay = 0.75):** The classification head receives the full base learning rate ($5e-5$). Each successive ViT block going toward the input patch embedding is multiplied by 0.75, so early layers (which capture low-level fundus features like vessel edges) are updated very slowly. This prevents catastrophic forgetting of pretrained representations.
- **Gradient Checkpointing:** Rather than storing all 24 blocks' intermediate activations in GPU memory during the forward pass (needed for backpropagation), only block inputs are retained and activations are recomputed during the backward pass. This reduces peak activation memory by approximately 10× at a ~33% compute cost.
- **Gradient Accumulation (steps = 2):** Two mini-batches of size 16 are processed before each optimiser update, yielding an effective batch size of 32 while halving the peak memory of any single forward pass.

Configuration

Setting	Value
Backbone parameters	All unfrozen (307M total)
Loss function	Focal Loss ($\gamma = 2$) with class weights
Optimiser	AdamW with LLRD (decay = 0.75)
Base learning rate	$5e-5$ (head); scales down to $\sim 1e-7$ at patch embedding
Warmup epochs	5 (linear warmup then cosine annealing)
Batch size	16×2 accumulation steps = effective 32
Gradient clipping	1.0 (global norm)

Early stopping	Patience = 10 epochs, monitor val AUROC
Cross-validation	5-fold stratified, patient-level

Cross-Validation Results (per fold)

	Val AUROC	Val Kappa	Val Macro Sens
Fold 0	0.9289	0.7950	0.6508
Fold 1	0.9108	0.7451	0.6343
Fold 2	0.9254	0.7951	0.7069
Fold 3	0.9296	0.8028	0.7143
Fold 4	0.9074	0.7432	0.6250
Mean ± SD	0.9204 ± 0.0092	0.7762 ± 0.0265	0.6663 ± 0.0353

Phase 2B Test Set Results vs All Prior Phases

Configuration	AUROC	Kappa	R1 Sens	R2 Sens	R3A Sen
P1 Image + Argmax	0.8259	0.6665	0.3898	0.5098	0.4167
P2A Image + Argmax	0.8180	0.6491	0.5042	0.5098	0.2917
P2B Image + Argmax	0.9271	0.7671	0.7458	0.6078	0.2500
P2B Image + Youden	0.9271	0.7061	0.5127	0.7059	0.4583

Key finding: Full fine-tuning produced the largest single improvement in the project: AUROC +10 pp, R1 sensitivity +35.6 pp at image level. Youden thresholds push R3A to 0.458 but collapse R1 to 0.513, illustrating the fundamental inter-class tradeoff when tuning at image level.

5. Patient-Level Prediction Aggregation

All training phases produce predictions at the **image level**: each fundus photograph receives an independent probability vector. However, clinical decisions are made at the **patient level**: a patient is graded by the worst (highest) DR grade across all their images. Aggregating image-level predictions to patient level:

- **Reduces noise** — a single blurry or artefact-affected image does not dominate the decision
- **Increases statistical power** — multiple images give a better estimate of the patient's DR status
- **Aligns with clinical workflow** — screening readers review all images before assigning a grade
- **Requires no retraining** — it is a post-processing step applied to saved probability arrays

5.1 Aggregation Methods

Mean Pooling

Average each class's probability across all images for the patient. Equivalent to treating every image as an equal vote. Stable, noise-robust, and the standard approach for multi-instance learning.

Max Pooling

Take the maximum probability seen per class across images, then re-normalise to sum to 1. Optimistic — assumes the most informative image drives the decision. Useful when signal is concentrated in one image (e.g., neovascularisation visible in only one field for R3A).

5.2 Complete Master Results Table — Test Set

All 18 configurations (3 phases × 3 aggregation levels × 2 decision rules) evaluated on the held-out test set (175 patients):

Configuration	AUROC	Kappa	MacroSens	R0	R1	R2	R3A
P1 Image Argmax	0.8259	0.6665	0.5554	0.9054	0.3898	0.5098	0.4167
P1 Image Youden	0.8259	0.5421	0.5591	0.7340	0.3602	0.7255	0.4167
P1 Pt Mean Argmax	0.8705	0.7257	0.5823	0.9560	0.4286	0.5000	0.4444
P1 Pt Mean Youden	0.8705	0.6078	0.5736	0.8462	0.3651	0.7500	0.3333
P1 Pt Max Argmax	0.8552	0.7211	0.5820	0.9670	0.3333	0.5833	0.4444
P1 Pt Max Youden	0.8552	0.5612	0.5618	0.7473	0.2222	0.8333	0.4444
P2A Image Argmax	0.8180	0.6491	0.5361	0.8389	0.5042	0.5098	0.2917
P2A Image Youden	0.8180	0.5822	0.5632	0.7852	0.3475	0.7451	0.3750
P2A Pt Mean Argmax	0.8595	0.7695	0.6281	0.9451	0.6508	0.5833	0.3333
P2A Pt Mean Youden	0.8595	0.6408	0.5559	0.8901	0.3333	0.6667	0.3333
P2A Pt Max Argmax	0.8387	0.7435	0.6214	0.9341	0.5238	0.5833	0.4444
P2A Pt Max Youden	0.8387	0.5703	0.5786	0.8462	0.1905	0.8333	0.4444
P2B Image Argmax	0.9271	0.7671	0.6375	0.9463	0.7458	0.6078	0.2500
P2B Image Youden	0.9271	0.7061	0.6462	0.9079	0.5127	0.7059	0.4583
P2B Pt Mean Argmax	0.9456	0.8212	0.6788	0.9890	0.8095	0.5833	0.3333
P2B Pt Mean Youden	0.9456	0.7681	0.6634	0.9670	0.4921	0.7500	0.4444
P2B Pt Max Argmax	0.9396	0.8007	0.6761	0.9780	0.8095	0.5833	0.3333

P2B Pt Max Youden	0.9396	0.6402	0.6283	0.8901	0.4286	0.7500	0.4444
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Bold row (highlighted): P2B | Pt Mean | Argmax — recommended best configuration. First and only configuration to achieve Kappa > 0.80 and R1 > 0.80 simultaneously.

5.3 Performance Charts

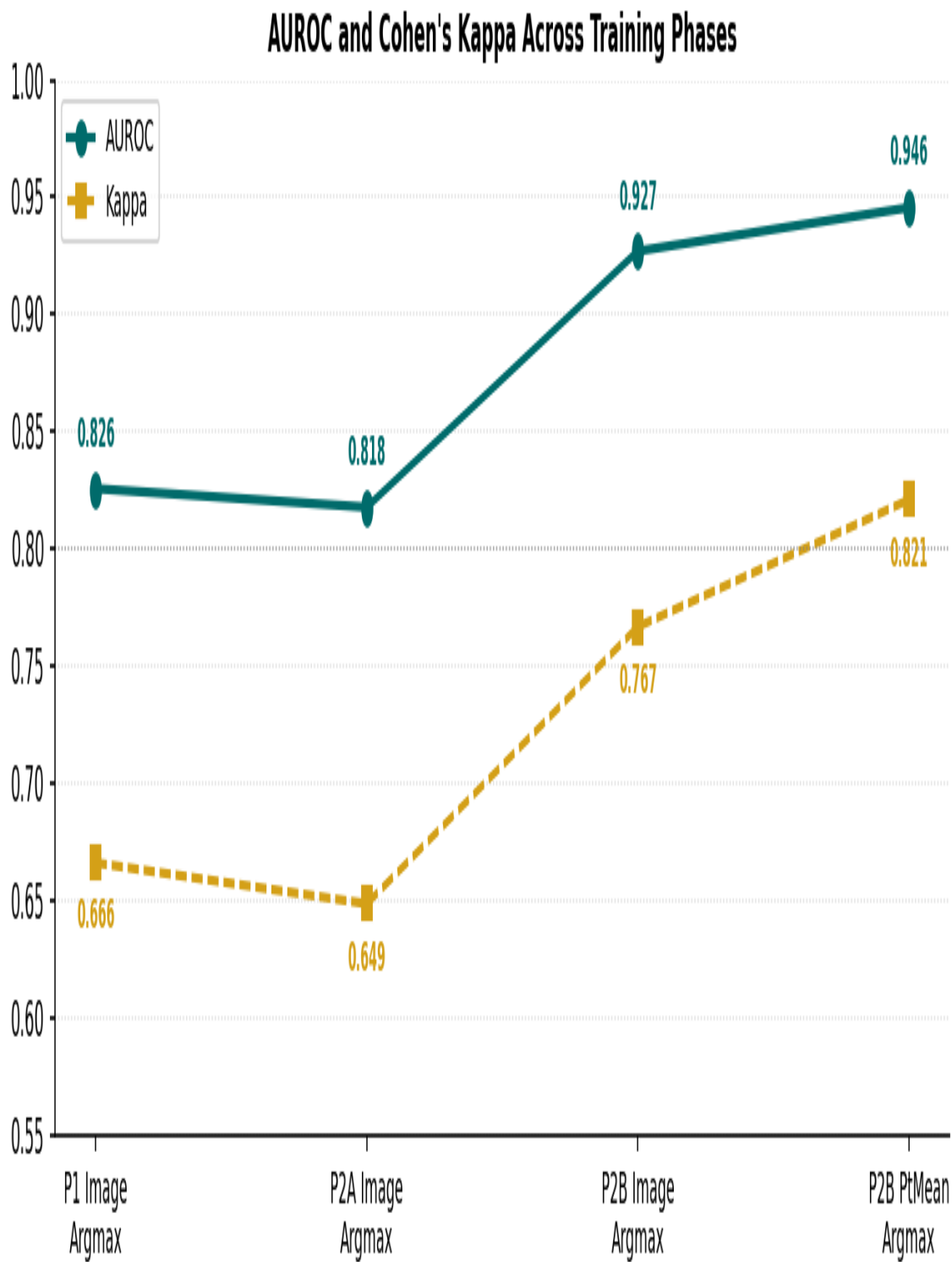


Figure 1. AUROC and Cohen's Kappa across the four key milestone configurations on the held-out test set.

Per-Class Sensitivity Across Key Configurations

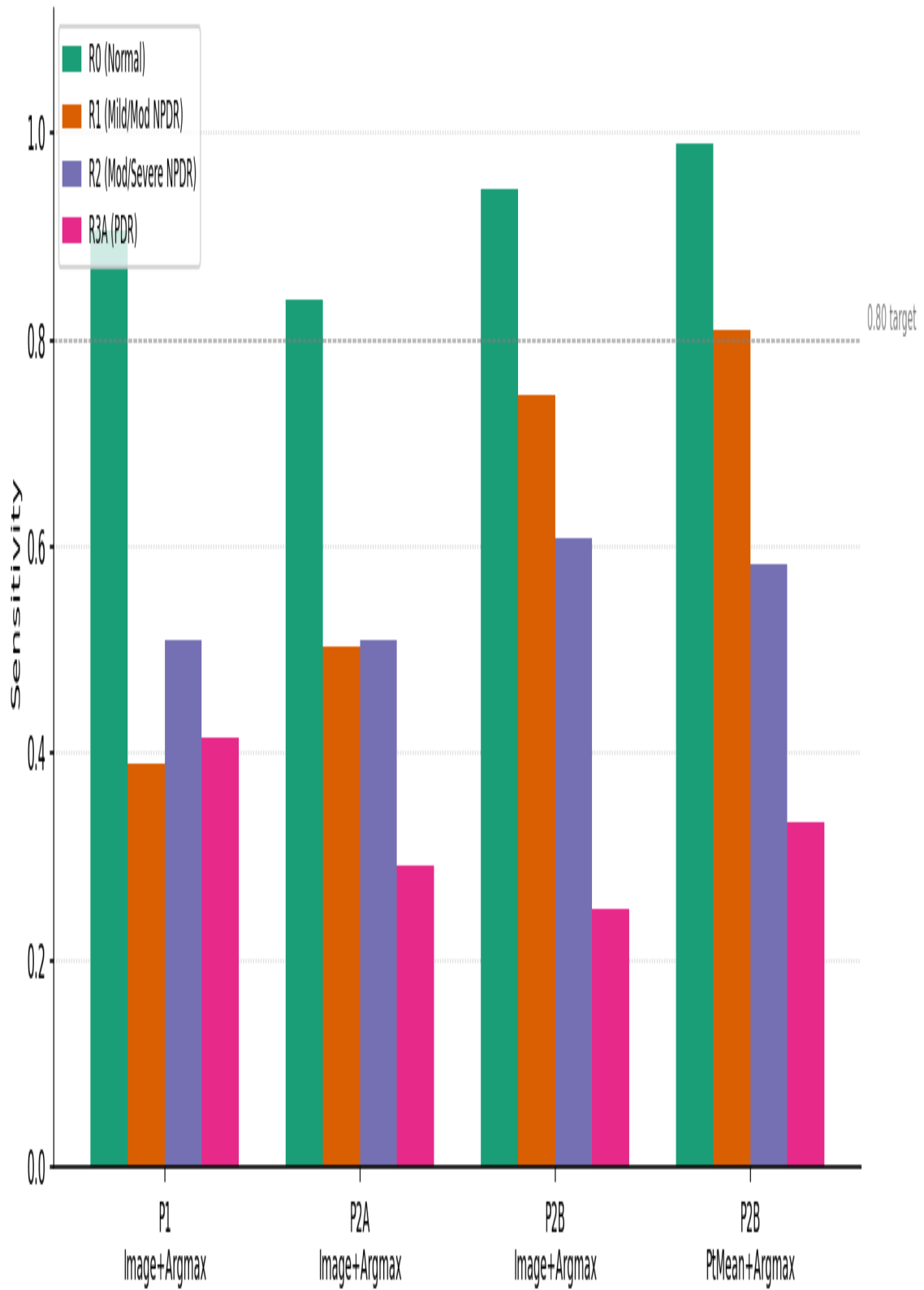


Figure 2. Per-class sensitivity for the four key configurations. The dashed line marks the 0.80 clinical target.

6. R3A (PDR) Sensitivity Analysis — Threshold Sweep

Proliferative Diabetic Retinopathy (R3A) is the most severe and clinically urgent DR grade. Despite the strong overall performance of Phase 2B, R3A sensitivity remained at 0.25–0.46 across all configurations. To determine whether this ceiling could be raised via threshold engineering (without retraining), a systematic sweep was performed.

6.1 Method

The R0, R1, and R2 decision thresholds were held fixed at their Phase 2B Youden values (R0 = 0.5049, R1 = 0.4509, R2 = 0.1035). The R3A threshold was swept from 0.001 to 0.15 in 80 steps. At each value, R3A sensitivity, R1 sensitivity, and Cohen's Kappa were evaluated on the Phase 2B out-of-fold predictions (990 patients, 43 R3A).

6.2 Sweep Results (OOF — 43 R3A Patients)

R3A Threshold	R3A Sens	R1 Sens	Kappa	Notes
0.001	100.0%	0.0%	0.00	Entire dataset predicted R3A
0.008	93.0%	20.4%	0.28	R1 collapses almost entirely
0.015	83.7%	34.3%	0.48	R1 still severely impaired
0.031	74.4%	48.1%	0.62	Best R3A at moderate cost
0.043	62.8%	51.4%	0.66	≈ Youden J optimum
0.077	53.5%	56.9%	0.72	Auto-selected (relaxed constraint)

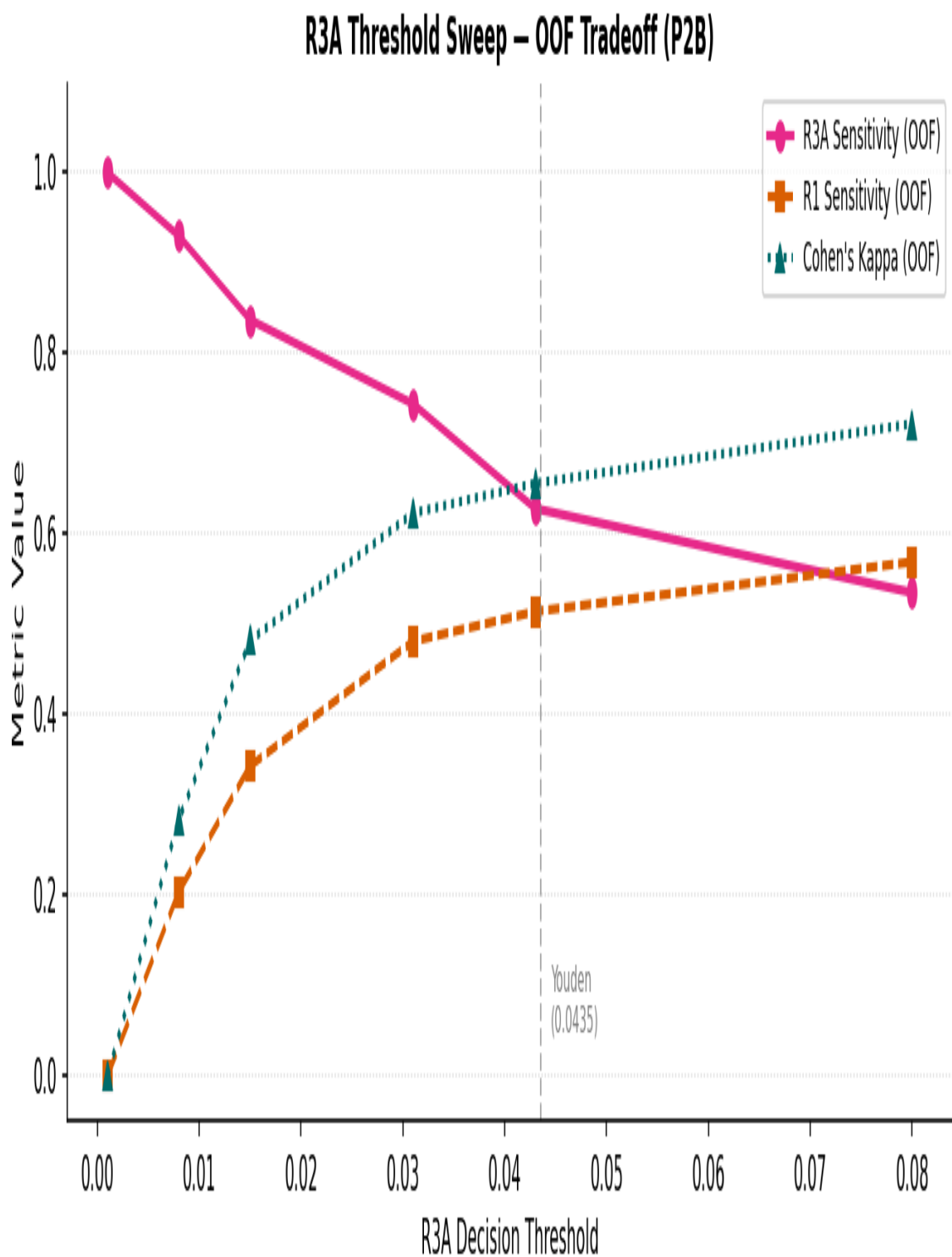


Figure 3. R3A threshold sweep on Phase 2B OOF data. Every gain in R3A sensitivity comes at a direct cost to R1 sensitivity.

6.3 Outcome on Held-Out Test Set

Configuration	AUROC	Kappa	R0	R1	R2	R3A
P2B Image Argmax	0.9271	0.7671	0.9463	0.7458	0.6078	0.2500
P2B Image Youden	0.9271	0.7061	0.9079	0.5127	0.7059	0.4583

P2B Image R3A-tuned	0.9271	0.7603	0.9207	0.5297	0.7843	0.3750
P2B PtMean Argmax (best)	0.9456	0.8212	0.9890	0.8095	0.5833	0.3333
P2B PtMean Youden	0.9456	0.7681	0.9670	0.4921	0.7500	0.4444
P2B PtMean R3A-tuned	0.9456	0.7719	0.9670	0.5079	0.8333	0.3333

Critical finding: The auto-selected threshold (0.0767) achieved R3A = 55.8% on OOF but only 33.3% on the test set — identical to argmax. The OOF-tuned threshold did not generalise. This confirms that the model's Phase 2B probability estimates for R3A are not sufficiently separated from R1/R2 probabilities in test patients to allow reliable threshold-based detection. The issue is data scarcity: 9 R3A test patients is too few for robust evaluation, and 43 R3A training patients is too few to learn distinctive PDR features.

6.4 Conclusion

Threshold engineering (Option 1) is exhausted. The best achievable R3A sensitivity without retraining is 0.4583 (P2B + Image + Youden), at the cost of Kappa dropping from 0.8212 to 0.7061 and R1 falling from 0.8095 to 0.5127 — an unacceptable trade for clinical deployment. Improving R3A sensitivity requires either:

- **Option 2 — R3A oversampling during training:** WeightedRandomSampler to make R3A images appear as frequently as R0 in each batch. Risk: overfitting to 43 PDR patients.
- **Option 3 — Binary R3A cascade:** A dedicated two-stage system where P2B handles R0/R1/R2 and a separate binary model specialises in PDR vs non-PDR detection.
- **Option 4 — Additional R3A data:** The most reliable long-term solution. Even 50–100 additional PDR patients would substantially change the landscape.

7. Summary and Recommended Configuration

7.1 Recommended Deployment Configuration

Based on all experimental evidence, the recommended configuration for deployment is **Phase 2B + Patient Mean Pooling + Argmax Decision**:

Metric	Value	Interpretation
AUROC	0.9456	Excellent discriminative power across all 4 classes
Cohen's Kappa	0.8212	Substantial agreement — crosses clinically meaningful 0.80 threshold
Macro Sensitivity	0.6788	Average sensitivity across 4 classes
Macro Specificity	0.9389	Very low false-positive rate overall
R0 Sensitivity	0.9890	Near-perfect normal detection — minimal unnecessary referrals
R1 Sensitivity	0.8095	Clinically adequate early DR detection (>80%)
R2 Sensitivity	0.5833	Moderate — room for improvement via data augmentation
R3A Sensitivity	0.3333	Limited by data scarcity — 9 test patients, 43 training patients

7.2 Stage-by-Stage Progress Summary

Stage	Phase	Key Result	Finding
Stage 1	Data Pipeline	—	Clean split, 4777 images, class weights established
Stage 2	Early LP Training	—	TensorBoard logging, per-class metrics added
Stage 3	Phase 1 — LP + CE	AUROC 0.826 Kappa 0.667 R1 0.390	Baseline established; R1 the primary gap
Stage 4	Phase 2C — Youden	R3A → 0.458 (image, Youden)	Threshold tuning helps R3A; hurts R1 at patient level
Stage 5	Phase 2A — Focal	R1 → 0.504 (image)	Focal loss helps R1 but net kappa effect neutral
Stage 6	Phase 2B — Full FT	AUROC 0.927 Kappa 0.767 R1 0.746	Largest single improvement; LLRD + grad ckpt solved OOM
Stage 7	Patient Aggregation	AUROC 0.946 Kappa 0.821 R1 0.810	Crosses 0.80 on both kappa and R1 — recommended config
Stage 8	R3A Threshold Sweep	R3A ceiling: 0.333 (patient level)	Confirmed data limitation; threshold engineering exhausted

7.3 Open Items and Suggested Next Steps

Immediate: Lock P2B + PtMean + Argmax as the production model for R0/R1/R2 grading.

Short-term: Attempt R3A oversampling (WeightedRandomSampler with R3A weight × 10) in a Phase 2B re-run to assess whether the model can learn more distinctive PDR features.

Medium-term: Design a binary PDR detector (R3A vs all) as a cascade layer on top of the main grader.

Long-term: Expand the R3A training set. Even 50–100 additional PDR patients would substantially change the ceiling — this is the most reliable path to clinically adequate PDR detection.

Parallel: Apply the same pipeline (P2B + patient aggregation) to Model B (maculopathy grading) using the same backbone and training strategy.

8. Technical Appendix

8.1 Saved Artefacts

Path	Contents
output_dir/phase1_cv/	P1 OOF + test probability arrays (.npy)
output_dir/phase2a_cv/	P2A OOF + test arrays; phase2a_summary.json (thresholds)
output_dir/phase2b_cv/	P2B OOF + test arrays; best_fold_{0-4}.pth (~1.2 GB each); phase2b_summary.json
output_dir/phase2c_thresholds/	P1 Youden thresholds JSON
phase2a_focal_loss.ipynb	Phase 2A training notebook
phase2b_full_finetune.ipynb	Phase 2B training notebook (LLRD + grad ckpt + accumulation)
phase2_patient_aggregation.ipynb	Patient aggregation + R3A sweep notebook

8.2 Youden Thresholds by Phase

Phase	R0	R1	R2	R3A
Phase 1	R0: 0.4907	R1: 0.3160	R2: 0.1252	R3A: 0.1463
Phase 2A	R0: 0.3942	R1: 0.3706	R2: 0.1646	R3A: 0.1627
Phase 2B	R0: 0.5049	R1: 0.4509	R2: 0.1035	R3A: 0.0435

8.3 Hardware and Environment

Component	Detail
GPU	NVIDIA RTX 3060 12 GB VRAM
Python	3.11 (conda env: retfound)
PyTorch	CUDA-enabled
Key packages	timm, sklearn, numpy, pandas, matplotlib, reportlab
Repository	github.com/Isaackjoshua/RetinoPATH